IE206 Advanced Question Bank: Detailed Solutions

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1 Linear Regression

Q1. Laplace Noise and MLE Objective

Assume the model

$$y_i = \mathbf{x}_i^\top \beta + \varepsilon_i, \qquad \varepsilon_i \overset{i.i.d.}{\sim} \text{Laplace}(0, b).$$

The Laplace density is

$$f_\varepsilon(e) = \frac{1}{2b} \exp\left(-\frac{|e|}{b}\right).$$

Hence the likelihood is

$$L(\beta, b) = \prod_{i=1}^{n} \frac{1}{2b} \exp\left(-\frac{|y_i - \mathbf{x}_i^\top \beta|}{b}\right),$$

and the negative log-likelihood is

$$-\log L(\beta, b) = n \log(2b) + \frac{1}{b} \sum_{i=1}^{n} |y_i - \mathbf{x}_i^\top \beta|.$$

For fixed b, MLE for β is

$$\hat{\beta}_{\text{MLE}} = \arg\min_{\beta} \sum_{i=1}^{n} |y_i - \mathbf{x}_i^\top \beta|,$$

not least squares. So under Laplace noise, *L1 regression* (LAD) is the correct MLE objective.

Robustness comparison:

- L2 loss penalizes residual r as r^2; outliers get very large influence.
- L1 loss penalizes residual as $|r|$; influence grows much slower.
- Therefore Laplace-MLE/L1 is more robust than Gaussian-MLE/L2 to heavy tails/outliers.

Q2. Perfect Collinearity

Let two features satisfy $\mathbf{x}_{\cdot,2} = c\,\mathbf{x}_{\cdot,1}$ exactly. Then columns of X are linearly dependent, so $\text{rank}(X) < p$ and

$$X^\top X \text{ is singular.}$$

OLS objective:

$$J(\beta) = \|y - X\beta\|_2^2.$$

If $v \in \mathcal{N}(X)$ (null space), then $X(\beta + v) = X\beta$, so

$$J(\beta + v) = J(\beta).$$

Hence minimizers are not unique: infinitely many β give the same fit.

Geometrically, moving along null directions does not change projection $X\beta$ in data space.

Ridge solves

$$\hat{\beta}_\lambda = \arg\min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2,$$

with closed form

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

For $\lambda > 0$, $X^\top X + \lambda I \succ 0$, so solution is unique. As $\lambda \to 0^+$, $\hat{\beta}_\lambda$ approaches the minimum-norm OLS solution $X^+ y$ (Moore–Penrose pseudoinverse solution).

Q3. High R^2 but Bad Test Error

Example construction:

- Training data: $x_i \in [-1, 1]$, $y_i = \sin(\pi x_i) + \epsilon_i$ with small noise.
- Fit a very high-degree polynomial (e.g., degree $n-1$ interpolation).

Training R^2 can be extremely high (often near 1), but test error is poor because the learned curve has high oscillation between training points (high variance).

Bias–variance view at a test point x:

$$\mathbb{E}\big[(Y - \hat{f}(x))^2\big] = \underbrace{\sigma^2}_{\text{noise}} + \underbrace{\big(\mathbb{E}[\hat{f}(x)] - f(x)\big)^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}(\hat{f}(x))}_{\text{variance}}.$$

Over-parameterized models can have low bias and high variance, so generalization degrades despite high training R^2.

Q4. Hat Matrix and Influence

For full-column-rank $X \in \mathbb{R}^{n \times p}$,

$$H = X(X^\top X)^{-1} X^\top.$$

We need $\sum_i h_{ii} = p$.

Use trace:

$$\sum_{i=1}^{n} h_{ii} = \mathrm{tr}(H) = \mathrm{tr}\left(X(X^\top X)^{-1} X^\top\right) = \mathrm{tr}\left((X^\top X)^{-1} X^\top X\right) = \mathrm{tr}(I_p) = p.$$

Interpretation: total leverage equals number of fitted parameters. So $\mathrm{tr}(H)$ behaves like model complexity (effective degrees of freedom).

Why high leverage points distort fit:

- $\hat{y} = Hy$, so $\hat{y}_i = \sum_j h_{ij} y_j$.
- Large h_{ii} means point i strongly controls its own fitted value.
- If such a point is extreme in feature space (and/or has large residual), it can rotate/shift the regression hyperplane disproportionately.

2 Logistic Regression

Q5. Convexity of Logistic Loss

Binary labels $y_i \in \{0, 1\}$, scores $z_i = \mathbf{x}_i^\top \beta$, probabilities $p_i = \sigma(z_i) = 1/(1 + e^{-z_i})$. Negative log-likelihood:

$$\ell(\beta) = \sum_{i=1}^{n} \left[\log(1 + e^{z_i}) - y_i z_i\right].$$

Gradient:

$$\nabla \ell(\beta) = X^\top (p - y).$$

Hessian:

$$\nabla^2 \ell(\beta) = X^\top W X, \quad W = \mathrm{diag}(p_i(1 - p_i)).$$

Each $p_i(1-p_i) \geq 0$, so for any v,

$$v^\top \nabla^2 \ell(\beta) v = (Xv)^\top W (Xv) = \sum_i w_i (Xv)_i^2 \geq 0.$$

Hence Hessian is PSD and logistic loss is convex.

Convexity implies every local minimum is global. If Hessian is PD on feasible directions (e.g., full rank plus no complete separation), minimizer is unique.

Q6. Decision Boundary and Feature Maps

Logistic model:

$$\mathbb{P}(y = 1 \mid x) = \sigma(\mathbf{w}^\top x + b).$$

Decision rule at threshold 0.5:

$$\sigma(\mathbf{w}^\top x + b) \geq 0.5 \iff \mathbf{w}^\top x + b \geq 0.$$

So boundary is linear in original features.

If we use transformed features $\phi(x)$,

$$\mathbb{P}(y = 1 \mid x) = \sigma(\mathbf{w}^\top \phi(x) + b).$$

Boundary is linear in ϕ-space but can be nonlinear in original x-space. So logistic regression remains a linear classifier in the chosen feature representation.

Q7. Extreme Class Imbalance (99% class 0)

Why accuracy misleads: Predicting always class 0 gives 99% accuracy but 0 recall for class 1.

Behavior of logistic regression: MLE emphasizes likelihood across all points; with severe imbalance, intercept tends strongly negative and predicted probabilities often stay below 0.5.

Decision threshold effect: Default threshold 0.5 usually yields many false negatives. In deployment one should tune threshold by costs/ROC/PR curve.

Optimal threshold under unequal costs: Let $\eta(x) = \mathbb{P}(Y = 1 \mid x)$.

- Cost of predicting 1: $C_{FP}\mathbb{P}(Y = 0 \mid x) = C_{FP}(1 - \eta)$.
- Cost of predicting 0: $C_{FN}\mathbb{P}(Y = 1 \mid x) = C_{FN}\eta$.

Predict class 1 when

$$C_{FP}(1 - \eta) \leq C_{FN}\eta \iff \eta \geq \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

Equivalent likelihood-ratio form:

$$\frac{p(x \mid 1)}{p(x \mid 0)} \geq \frac{C_{FP}}{C_{FN}} \frac{\pi_0}{\pi_1}.$$

3 Perceptron

Q8. Non-Separable Data

Perceptron update on mistakes:

$$\mathbf{w}_{t+1} = \mathbf{w}_t + y_i \mathbf{x}_i.$$

If algorithm stopped after finite steps, then no future mistakes occur, so final hyperplane correctly classifies all training points. That implies separability, contradiction. Therefore for non-separable data, perceptron keeps making mistakes forever.

Asymptotic behavior: weights generally do not converge to a fixed point; they keep moving among conflicting constraints, often with growing norm and oscillating direction.

Soft-margin SVM comparison:

- Perceptron optimizes no regularized convex objective in non-separable case.
- Soft-margin SVM solves convex optimization with slack, always well-posed and convergent.
- SVM returns a controlled margin/error tradeoff; perceptron does not.

Q9. Mistake Bound and Scaling

Classical bound for separable data:

$$T \leq \left(\frac{R}{\gamma}\right)^2,$$

where $R = \max_i \|\mathbf{x}_i\|$ and γ is geometric margin. This is equivalent to $T \leq (RB)^2$ with $B = 1/\gamma$.

Effects of scaling:

- Uniform scaling $\mathbf{x} \mapsto a\mathbf{x}$ scales both R and γ by a, so ratio R/γ unchanged.
- Non-uniform scaling can increase R and shrink effective margin, worsening bound.
- Feature normalization tends to control R and improve practical convergence speed.

Can bad scaling cause divergence?

- If data is linearly separable, perceptron still converges theoretically (finite mistakes), though may be extremely slow numerically.
- For non-separable data, it does not converge regardless of scaling.

4 Support Vector Machines

Q10. Hard vs Soft Margin Limits

Soft-margin primal:

$$\min_{\mathbf{w},b,\xi\geq 0} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n} \xi_i, \quad y_i(\mathbf{w}^\top x_i + b) \geq 1 - \xi_i.$$

If data is separable, there exist feasible points with $\xi_i = 0$. As $C \to \infty$, any nonzero slack becomes very expensive, so optimizer approaches zero-slack solution, i.e. hard-margin SVM.

As $C \to 0$, slack is nearly free, objective mainly minimizes $\|\mathbf{w}\|^2$, so $\mathbf{w} \to 0$ (trivial classifier tendency), with many margin violations tolerated.

Q11. KKT Interpretation for Soft-Margin SVM

Dual constraints: $0 \leq \alpha_i \leq C$. KKT:

$$\alpha_i\big(y_i f_i - 1 + \xi_i\big) = 0, \quad (C - \alpha_i)\xi_i = 0, \quad f_i = \mathbf{w}^\top x_i + b.$$

Cases:

1. $\alpha_i = 0$: then $\xi_i = 0$ and $y_i f_i \geq 1$ (typically outside margin). Not a support vector.
2. $0 < \alpha_i < C$: then $\xi_i = 0$ and $y_i f_i = 1$. Exactly on margin (support vector).
3. $\alpha_i = C$: on/inside margin. Usually $y_i f_i \leq 1$; if $y_i f_i < 1$ then inside margin, and if $y_i f_i < 0$ point is misclassified.

Q12. Margin Geometry

Canonical separating hyperplanes:

$$\mathbf{w}^\top x + b = 1 \quad \text{and} \quad \mathbf{w}^\top x + b = -1.$$

Distance between parallel hyperplanes $\mathbf{w}^\top x + b = c_1$ and c_2 is

$$\frac{|c_1 - c_2|}{\|\mathbf{w}\|}.$$

Hence margin width is

$$\frac{|1 - (-1)|}{\|\mathbf{w}\|} = \frac{2}{\|\mathbf{w}\|}.$$

So maximizing margin is equivalent to minimizing $\|\mathbf{w}\|$, or equivalently minimizing $\frac{1}{2}\|\mathbf{w}\|^2$.

Q13. Why the Dual Uses Inner Products; PSD Requirement

Soft/hard SVM dual objective has form

$$\max_\alpha \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle,$$

so data appears only through pairwise inner products. Replacing $\langle x_i, x_j \rangle$ by $K(x_i, x_j)$ gives kernel SVM.

If K is not PSD, Gram matrix can be indefinite. Then the dual is not guaranteed concave (optimization may become non-convex with local optima), and no valid RKHS/Mercer interpretation is guaranteed.

5 KNN and Bayes

Q14. Curse of Dimensionality Distance Concentration

Let $X, Y \sim \text{Unif}([0,1]^d)$ independent. Define

$$D^2 = \|X - Y\|_2^2 = \sum_{j=1}^{d} (X_j - Y_j)^2.$$

Each term is bounded and i.i.d. with mean $1/6$. By law of large numbers,

$$\frac{D^2}{d} \to \frac{1}{6} \quad \text{(in probability)}.$$

So pairwise distances concentrate around $\sqrt{d/6}$. Spread is order $\sqrt{d}$, mean is order d in squared distance, so relative spread shrinks as $1/\sqrt{d}$.

Therefore nearest and farthest neighbor distances become similar in relative terms, i.e.

$$\frac{D_{\max} - D_{\min}}{D_{\min}} \to 0,$$

which breaks "nearest means similar" intuition and hurts KNN.

Q15. Bayes Error vs 1-NN; Bayes Consistency

At a point x, let $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$.

Bayes conditional error:

$$R^*(x) = \min\{\eta(x), 1 - \eta(x)\}.$$

As sample size grows, the nearest neighbor label behaves like an independent Bernoulli draw with parameter $\eta(x)$. Then 1-NN conditional error tends to

$$R_{1NN}(x) = 2\eta(x)(1 - \eta(x)).$$

Since $2\eta(1 - \eta) \leq 2\min\{\eta, 1 - \eta\}$,

$$R_{1NN}(x) \leq 2R^*(x).$$

This gives the standard bound intuition.

Why $k \to \infty$ with $k/n \to 0$ is Bayes-consistent:

- $k \to \infty$: enough neighbors to average labels, reducing variance.
- $k/n \to 0$: neighbor radius shrinks to 0, so local class probability estimate is asymptotically unbiased.

Hence risk converges to Bayes risk.

Q16. Scaling Trap Construction

Let training set (for 1-NN):

$$(0, 1000) \to 0,\ (0, 900) \to 0,\ (1, 0) \to 1,\ (1, 100) \to 1.$$

Query: $q = (1, 950)$ (informative feature is first coordinate, so true class should be 1).

Without normalization:

$$d(q, (0, 900)) = \sqrt{1 + 50^2} \approx 50.01,$$

$$d(q, (1, 100)) = 850,$$

so nearest neighbor is class 0 (wrong).

After min-max normalization per feature:

$$x_1 \in [0, 1], \quad x_2 \in [0, 1000],$$

query becomes $(1, 0.95)$, and

$$d((1, 0.95), (1, 0.1)) = 0.85,$$

$$d((1, 0.95), (0, 0.9)) \approx 1.001,$$

so nearest is class 1 (correct). This shows distance domination by large-scale irrelevant features.

6 Decision Trees and Random Forest

Q17. Perfect Feature and Information Gain

Information gain for feature X:

$$IG(Y, X) = H(Y) - H(Y \mid X).$$

If X perfectly determines Y, then each split child is pure, so

$$H(Y \mid X) = 0.$$

Therefore

$$IG(Y, X) = H(Y).$$

So gain equals initial entropy.

Q18. Why Unrestricted Trees Have Infinite VC Dimension

Take any m distinct real points $x_1 < \cdots < x_m$ on $\mathbb{R}$. A deep enough axis-aligned tree can place split thresholds between consecutive points to isolate each point in its own leaf. Then assign each leaf label 0 or 1 arbitrarily. So every labeling of these m points is realizable. Since m is arbitrary, VC dimension is infinite.

Q19. Why Bagging Reduces Variance but Not Bias

Let $\hat{f}_b(x)$ be predictor from bootstrap sample b, and ensemble

$$\bar{f}_M(x) = \frac{1}{M} \sum_{b=1}^{M} \hat{f}_b(x).$$

For squared loss:

$$\text{MSE} = \underbrace{\left(\mathbb{E}[\bar{f}_M] - f\right)^2}_{\text{bias}^2} + \underbrace{\text{Var}(\bar{f}_M)}_{\text{variance}} + \sigma^2.$$

Averaging mostly affects variance. If base learners have variance σ_f^2 and pairwise correlation ρ,

$$\text{Var}(\bar{f}_M) = \rho\sigma_f^2 + \frac{1-\rho}{M}\sigma_f^2.$$

As M grows, second term vanishes, first remains. Bias comes from model class and is not fundamentally removed by averaging.

Q20. Why Small Feature Subset Size in RF Reduces Correlation

If all trees can always use strongest predictors, many trees split similarly, increasing correlation ρ. Random feature subsampling forces diverse split choices, decreasing ρ in

$$\text{Var}(\bar{f}_M) = \rho\sigma_f^2 + \frac{1-\rho}{M}\sigma_f^2.$$

Lower ρ lowers ensemble variance substantially. Tradeoff: very small subset size may weaken each tree (slightly higher bias).

7 Model Evaluation

Q21. Accuracy Paradox Example

Confusion matrix (1000 samples, 5% positives):

	$\hat{y}=1$	$\hat{y}=0$
$y=1$	0	50
$y=0$	0	950

Accuracy $= (950+0)/1000 = 95\%$. Model is useless for positive class (recall $= 0$).

Q22. Higher Accuracy but Lower F1

Assume 1000 samples, 100 positives.

Model A:

$$TP = 10,\ FP = 0,\ FN = 90,\ TN = 900.$$

Accuracy $= 91\%$,

$$F1_A = \frac{2 \cdot 1 \cdot 0.1}{1+0.1} = 0.1818.$$

Model B:

$$TP = 70,\ FP = 120,\ FN = 30,\ TN = 780.$$

Accuracy $= 85\%$,

$$\text{Precision} = \frac{70}{190} = 0.3684, \quad \text{Recall} = 0.7, \quad F1_B \approx 0.4828.$$

So B has lower accuracy but much better F1 and is preferable when catching positives matters.

Q23. LOOCV and High Leverage

Given

$$\hat{y}_i^{(-i)} = \frac{\hat{y}_i - h_{ii} y_i}{1 - h_{ii}}.$$

Then LOOCV residual is

$$e_i^{(-i)} = y_i - \hat{y}_i^{(-i)} = \frac{y_i - \hat{y}_i}{1 - h_{ii}} = \frac{e_i}{1 - h_{ii}}.$$

So LOOCV squared error term scales as

$$\left(e_i^{(-i)}\right)^2 = \frac{e_i^2}{(1 - h_{ii})^2}.$$

If $h_{ii} \approx 1$, denominator is tiny and contribution explodes. This is why highly leveraged points can dominate LOOCV error.

8 Multiclass Classification

Q24. One-vs-All Failure Case

Construct 3-class data in $\mathbb{R}^2$:

- Class A near $(-2, 0)$, many points.
- Class B near $(2, 0)$, many points.
- Class C near $(0, 0)$, very few points.

For OvA classifier of class C (C vs A+B), negatives dominate and are on both sides of C; with linear model and imbalance, optimal score for C can be negative almost everywhere. Then argmax over OvA scores never predicts C, so the class disappears.

Why joint multiclass SVM helps: it learns all class scores together with constraints

$$\mathbf{w}_{y_i}^\top x_i - \mathbf{w}_j^\top x_i \geq 1 - \xi_i, \quad \forall j \neq y_i,$$

forcing C examples to beat all other classes by margin. This coupling prevents trivial elimination of a class.

Q25. All-Pairs (OvO) Complexity

For k classes:

- OvA: train k classifiers.
- OvO: train $k(k-1)/2$ classifiers.

So training/model storage is $O(k^2)$ for OvO.

When infeasible: large k (e.g., $k = 1000$ gives 499,500 classifiers; $k = 10{,}000$ gives about 50 million).

When OvO may be better than OvA:

- each binary problem is smaller and often better balanced,
- pairwise boundaries may be simpler,
- can improve accuracy despite more models.

Prediction cost is also heavier for OvO (often $O(k^2)$ voting, unless structured decoding like DAG is used).

9 Very Advanced

Q26. Bias–Variance for Regression; Why Classification is Harder

Let

$$Y = f(x) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid x] = 0, \quad \mathrm{Var}(\varepsilon \mid x) = \sigma^2.$$

For estimator $\hat{f}_D(x)$ (random over dataset D):

$$\mathbb{E}_{D,\varepsilon}\big[(Y - \hat{f}_D(x))^2 \mid x\big] = \sigma^2 + \big(\mathbb{E}_D[\hat{f}_D(x)] - f(x)\big)^2 + \mathrm{Var}_D(\hat{f}_D(x)).$$

This is exact decomposition for squared loss.

For classification with 0–1 loss, risk uses indicator $\mathbf{1}\{\hat{y} \neq y\}$, which is discontinuous/non-quadratic. There is no equally clean additive bias+variance decomposition in general. One typically uses surrogate losses, margin analysis, or approximate decompositions.

Q27. VC Dimension Comparison

- Linear classifiers in $\mathbb{R}^d$: VC $= d + 1$.
- Decision trees of depth k (axis-aligned, in $\mathbb{R}^d$): capacity grows exponentially with depth; typical bounds are on the order of $O(2^k \log d)$ (and at least $\Omega(2^k)$). Unrestricted depth gives infinite VC dimension.
- SVM with RBF kernel: without norm regularization, effective VC dimension is infinite (can shatter arbitrarily many points). Regularization controls effective capacity in practice.

Q28. Perceptron vs Logistic vs Hard SVM

Method	Loss / Objective	Margin behavior	Convergence / Robustness
Perceptron	Minimize mistakes implicitly via updates on misclassified points (non-convex objective surrogate)	No explicit max-margin objective; in separable case finds some separator	Converges only for separable data; unstable on non-separable; sensitive to scaling/outliers
Logistic regression	Convex log-loss $\sum \log(1 + e^{-yf(x)})$ (often with L2/L1 regularization)	Encourages large positive margins smoothly but not max-margin exactly	Global optimum (convex); probabilistic outputs; more robust optimization than perceptron
Hard-margin SVM	$\min \frac{1}{2}\|w\|^2$ s.t. $y_i f_i \geq 1$	Explicitly maximizes geometric margin	Convex QP with unique w for separable data; sensitive to label noise/outliers (no slack)

Bottom-line comparison. Perceptron is simplest but least robust outside separability. Logistic is probabilistic and convex. Hard SVM gives strongest geometric margin but requires clean separability; soft-margin SVM is the practical compromise.